
CHAPTER 3

Data engineering

A fundamental part of building any type of predictive model is the training data. This part details the process by which the data is obtained, processed, analysed and selected in order to feed the model with the most appropriate and necessary information to obtain better results both in training and in inference.

3.1 Source of data

Following an exhaustive search on the Internet, two datasets were finally used for the generation of the training data.

3.1.1. Football-Data

The first dataset was taken from the Football-Data website¹, which has data for all the matches of the major European and American leagues, as well as other more exotic European leagues, such as the Greek, Finnish or Swiss leagues. It contains results, statistics, bookmaker odds and other features such as the date of the match. For this analysis, data from the five major leagues - English, Spanish, Italian, German and French - from the 1993-1994 season to the 2021-2022 season was used. Table 3.3 shows the main attributes available for each party in this dataset. Attributes with the * symbol are not available in all leagues, and will therefore be omitted for the generation of the dataset. Some attributes omitted have been odds from other bookmakers and other types of betting odds, such as the Asian Handicap.

3.1.2. Wyscout

The data for the second data set, obtained from the work of Pappalardo et al., was provided to this paper by Wyscout, a world-leading company in the football industry [Pappalardo et al., 2019]. Data was collected by Wyscout using video expert analysis, called *operators*, one for each team and an extra one supervising. They are trained with a football data collection with a proprietary software *tagger*. Tagging consists of three phases [Pappalardo et al., 2019]:

1. **Setting formations:** an *operator* lists the starting line-ups and the positions and numbers of players on the field.

¹<https://www.football-data.co.uk/>

2. **Event tagging:** for each ball touch, *operator* selects a player and creates a new event in the timeline. Then, it adds an event type and a subtype. Finally it adds the coordinates in the event field and some additional attributes.
3. **Quality control:** two checks are carried out. The first one is automatic, an algorithm matches the events tagged by the two *operators* to validate that the events makes sense. The second check is manual.

This database contains all matches and their events from the five major European leagues (English, Spanish, German, Italian and French) during the 2017-2018 season, as well as from the World Cup 2018 and European Cup 2016. Table 3.1 shows the total amount of matches, events and players available.

competition	#matches	#events	#players
LaLiga (Spain)	380	628,659	619
Premier League (UK)	380	643,150	603
Serie A (Italy)	380	647,372	686
Bundesliga (Germany)	306	519,407	537
Ligue 1 (France)	380	632,807	629
World Cup 2018	64	101,759	736
European Cup 2016	51	78,140	552
	1,941	3,251,294	4,299

Table 3.1: List of competitions with their total number of matches, events and players.

This dataset is stored in multiple JSON files. For each competition there is a JSON file for the matches and another one for the match events. There are other JSON files that store information about players, teams, coaches and competitions. Table 3.4 shows the match features and events, which are the most remarkable datasets in this data base.

type	subtype	tags
pass	cross, simple pass	accurate, not accurate, key pass, opportunity, assist, goal
shot		accurate, not accurate, block, opportunity, assist, goal
duel	air duel, dribbles, tackles, ground loose ball	accurate, not accurate
free kick	corner, shot, goal kick, throw in, penalty, simple kick	accurate, not accurate, key pass, opportunity, assist, goal
offside touch	acceleration, clearance, simple touch	counter attack, dangerous ball lost, missed ball, interception, opportunity, assist, goal
foul		no card, yellow, red, 2nd yellow

Table 3.2: Event types, subevent types and tags en el Wyscout dataset.



Football-Data		
attribute	type	description
Div	string	League division
Date	string	Match date (dd/mm/yy)
HomeTeam	string	Home team
AwayTeam	string	Away team
FTHG	integer	Full Time Home Team Goals
FTAG	integer	Full Time Away Team Goals
FTR	string	Full Time Result (H=Home Win, D=Draw, A=Away Win)
HTHG	integer	Half Time Home Team Goals
HTAG	integer	Half Time Away Team Goals
HTR	string	Half Time Result (H=Home Win, D=Draw, A=Away Win)
Referee*	string	Match referee
HS	integer	Home Team Shots
AS	integer	Away Team Shots
HST	integer	Home Team Shots on Target
AST	integer	Away Team Shots on Target
HF	integer	Home Team Fouls Committed
AF	integer	Away Team Fouls Committed
HC	integer	Home Team Corners
AC	integer	Away Team Corners
HY	integer	Home Team Yellow Cards
AY	integer	Away Team Yellow Cards
HR	integer	Home Team Red Cards
AR	integer	Away Team Red Cards
B365H	float64	Bet365 home win odds
B365D	float64	Bet365 draw odds
B365A	float64	Bet365 away win odds
BSH	float64	Blue Square home win odds
BSD	float64	Blue Square draw odds
BSA	float64	Blue Square away win odds
BWH	float64	Bet&Win home win odds
BWD	float64	Bet&Win draw odds
BWA	float64	Bet&Win away win odds
WHH	float64	William Hill home win odds
WHD	float64	William Hill draw odds
WHA	float64	William Hill away win odds

Table 3.3: Attributes of a match in Football-Data dataset.

Matches		
attribute	type	description
wyId	integer	identifier of the match
competitionId	integer	identifies the competition on the <i>competition</i> document.
dateutc	string	date and time with format YYYY-MM-DD hh:mm:ss
roundID	integer	match-day of the competition
winner	integer	id of the team that won the game, or 0 if ended with a draw
label	string	name of the two clubs and the result of the match
venue	string	stadium where the match was held
dateutc	string	date and time with format YYYY-MM-DD hh:mm:ss
teamsData	dictionary	several subfields describing information about each team
teamsData subfields (for each team):		
hasFormation	integer	0 if no formation, and 1 otherwise
score	integer	number of goals scored by the team during the match
side	string	“home” or “away”
teamId	integer	identifier of the team
coachId	integer	identifier of the coach
bench	list	list of players that started on the bench and some basic statistics about the performance (goals, own goals, cards)
lineup	list	list of players that started on the lineup and some basic statistics about the performance (goals, own goals, cards)
substitutions	list	list of team’s substitutions during the match, describing the players involved and the minute

Events		
attribute	type	description
eventId	integer	identifier of the event’s type (Table 3.2)
subEventId	integer	identifier of the event’s subtype (Table 3.2)
eventName	string	name of the event’s type
subEventName	string	name of the subevent’s type
tags	list	list of event tags, additional information about the event (Table 3.2)
eventSec	integer	time when the event occurs (in seconds)
id	integer	unique identifier of the event
matchId	integer	identifier of the match the event refers to
matchPeriod	string	“1H” (first half of the match), “2H” (second half of the match)
positions	list	origin and destination positions associated with the event. Each position is a pair of coordinates (x, y).
playerId	integer	identifier of the player who generated the event
matchId	integer	identifier of the match the event refers to
teamId	integer	identifier of the player’s team

Table 3.4: Attributes of matches and events on original Wyscout datasets.



3.2 Dataset creation

In order to create a suitable dataset to train the model, different transformations had to be applied, the data from the databases had to be cleaned and processed and merged to obtain a more complete database. In total we have created three databases:

1. **Betting odds dataset:** created to train our baseline model. It consists of the Bet365 odds of the 2017-18 season matches of the five major European leagues.
2. **Wyscout based dataset:** contains the matches of the 2017-18 season of the five main European leagues. Each sample will correspond to a match, and 20 attributes represent the statistics corresponding to the last n matches by the players of each team's line-up. The features will be obtained either directly from the Wyscout dataset or generated from Wyscout events or statistics.
3. **Historical database:** contains data from the five major European leagues from the 1993-1994 season to the 2021-22 season. Each match will contain general statistics taken from the Football-Data dataset.

The main problem in the creation of the datasets has been the linking of both databases as their primary keys do not match, nor does the format of the team names match and, therefore, there was no straightforward way to connect the two databases. The solution that was proposed is an iterative solution. In the first iteration the team names that match in both datasets are detected, in the second iteration with the help of the Python library *difflib* the club names that are very similar in both datasets are detected, and finally the remaining teams (a dozen) were changed manually. Thus we could now associate matches from both datasets, as we can now identify in the Football-Data (Table 3.3) dataset the name of a club with its id in Wyscout (Table 3.4).

When matching both databases only the names of the teams have been considered as we are only interested in the matches of the 2017-2018 season, which are the ones that appear in Wyscout (Table 3.1).

dataset	#samples	#attributes
Betting odds dataset	1826	3
Wyscout based dataset	1826	20
Historical database	45383	29

Table 3.5: Number of samples and attributes of the datasets.

3.2.1. Wyscout based dataset

In this generated dataset, all samples are independent due to the transformations performed on the original dataset and the generated attributes that eliminate the temporal dependency between samples. The generated attributes have in common that they are all generated from features of the last n matches within a season. Therefore, each sample (match) already contains all the past information necessary to predict its outcome, actually it does not contain any features other than those of the past matches. This removes the temporal dependency between samples, making them independent.

The previous 4 matches have been used for the calculation of the attributes of each sample. It was decided to use this number of matches in order not to further reduce the

limited number of samples that are available, since by using the last 4 matches, the first 4 matches of every season of each team must be used, thus losing quite a few records. Specifically, with 4 matches we reduce from 1826 to 1414 samples. Moreover, in previous works with a similar approach, the best results have been obtained with values between 4 and 7 matches [Ulmer et al., 2013, Jain et al., 2021].

It has been questioned in the literature whether there is a relationship between teams' winning streaks and the outcome of matches, especially when the streak is one of victories [Heuer and Rubner, 2009]. Heuer and Rubner argue that a negative streak has a negative impact on the probability of winning, while a winning streak has no impact on a team's probability of winning. However, other studies contradict these claims and the intuition itself, showing how a winning streak results in a decrease in the probability of winning, and a losing streak in an increase in the probability of winning [Goddard, 2006]. The relationship between streaks and results remains uncertain.

In the generated dataset each sample contains 20 attributes, which will be described and discussed below:

- **mins_{n_S}²**: average number of minutes played by each team's line-up in the last n games. The values will therefore be between 0 and 90. This attribute should tell the model whether a team is playing with starters, substitutes or missing regular players from a team.
- **shots_{11S}**: absolute number of shots taken by each team's starter in the last n games.
- **shots_acc_{11S}**: average number of shots on target of the line-up in the last n games.
- **goals_S**: sum of goals scored by each team's starting players in the last n matches.
- **goals_ratio_S**: ratio of goals scored by the starting eleven over the total goals scored by the team in the last n games. This attribute will also be an indicator of whether important players are missing from the line-up or playing in the starting line-up.
- **passes_{11S}**: sum of passes made by each team's starting eleven in the last n matches.
- **passes_acc_{11S}**: passing success rate of each team's starting eleven in the last n matches.
- **keyPasses_S**: sum of key passes made by each team's starting eleven in the last n matches. Key passes are passes that create situations that result in a clear chance for the team.
- **attacking_S**: indicator of the attacking power of each team over the last n matches. It is calculated from the weighted average of the goals scored in the last n matches, in which the most recent match has more weight than the oldest.
- **defensa_S**: indicator of the defensive consistency of each team over the last n matches. It is calculated from the weighted average of goals conceded in the last n matches, where the most recent match carries more weight than the oldest.

It is important to point out that the statistics of a team in the last n matches only affect home and away matches. For example, in the Malaga vs Celta de Vigo, the sample will contain data from Malaga's last n home matches and Celta's last n away matches.

²Attribute with "S" have two versions, one for each side team: home ("H") and away ("A")

3.2.2. Historical dataset

The idea of this dataset is similar to that of [Wyscout based dataset](#), to create a dataset in which each sample is independent and contains as much information as possible about the match. The purpose of this database is to train this model with a larger number of matches and, therefore, be able to generalise better in the predictions. This will be discussed later in the Experiments section.

This dataset has been created only from the [Football-Data](#) database. In order to create it, data from the seasons between 1993 and 2021 from English Premier League, German Bundesliga, Spanish LaLiga and Italian Serie A have been used, and also data from French Ligue 1 since 1997 as no data from previous seasons was available.

This dataset was created by merging all available seasons from the five leagues and pre-processing the primary keys to match the Wyscout dataset. The team IDs in both datasets have been identified by mapping the team IDs from the Wyscout dataset to this dataset, and then the matching games have been identified, being those from the 2017-2018 season to which the ID from the Wyscout dataset has been mapped. For teams and matches that did not appear in the Wyscout dataset, a new unique ID has been generated. In total this dataset contains 45,383 matches, which is considerably larger than the Wyscout dataset.

As for the features that this dataset contains, all of them have been generated from the variables shown in [Table 3.3](#). Apart from the match metadata shown in variables such as competition, date, home and away team and result, statistics of the match such as shots, fouls or corners have been taken to create new variables for the dataset. The new variables that have been created are values that are intended to show the trend of a team in different time lags, i.e. for each team aggregations of these variables have been calculated in different time lags.

It is also worth mentioning that the source data did not include all data for all leagues and seasons, for example, we did not have information on the number of fouls committed in a match in the Spanish league until the 2005-2006 season, while in the English league we can know the number of fouls committed even in a match in the 2000-2001 season.

It is because of this difference in features between competitions and seasons that it has been decided to make different versions of the dataset with different variables and sizes, [Table 3.6](#) and [3.7](#) show the variables and the version of the dataset in which they appear.

As in the creation of the Wyscout dataset, the characteristics of this dataset bring together certain historical statistics of a team in a given time lag. This lag can be represented in two ways ([Table 3.7](#)). The first way is to use the last n matches of a team within the same season, which will be called from now on n_lag . The second way is to use a time interval such as the last d days, which will be called d_lag . It has been decided to use experimenting with d_lag to perform the lags, because it is possible to keep more matches from the beginning of each season. With the version of n_lag the first games of each season are lost, since it is calculated intra-season. However, d_lag is calculated inter-seasonally, so even if there is no data for a team for the last 2 months ($60D_lag$), there will certainly be data for the last year ($365D_lag$).

In the Wyscout dataset, where there were fewer samples (only one season), if large lags were made, many matches from the beginning of the season would be lost. On the con-

Feature	Basic	Basic w/ Derby	Complete	Basic Longterm	Basic Longt. w/ Derby	Comp. Longterm
Side (home/away)	X	X	X	X	X	X
Scored goals	X	X	X	X	X	X
Received goals	X	X	X	X	X	X
Points	X	X	X	X	X	X
Shots			X			X
Shots on target			X			X
Corners			X			X
Fouls			X			X
Yellow Cards			X			X
Red Cards			X			X
Derby features		X			X	
# features (columns)	80	104	200	56	80	140
# matches (rows)	32.997	23.267	21.635	17.373	13.536	11.895

Table 3.6: Historical database features and versions.

dataset	time characteristics	lags of time
Basic	Lags are done exactly in the same way as in the Wyscout dataset, taking the last n matches of a team	n_lag : Range from 1 to 10 previous matches.
Complete		
Basic Longterm	Streaks are represented by the matches played within the last d days.	d_lag : d is 15, 30, 60, 180, 365 (1 year), 730 (2 years) and 1825 (5 years) days.
Complete Longterm	Note: <i>Derby</i> features are represented with the last n matches.	

Table 3.7: Description of lag features.

trary, in this dataset, there are many more samples and, therefore, it has been decided to create longer lags, in order to give the model information not only about the recent performance, but also about the medium and long term performance. To achieve this, several versions of the dataset have been created where features have been added to show the performance of a team over a very long period of time, up to 5 years (1825 days).

In order to create the lags, the minimum required number of records for a particular lag to be valid has to be determined. It would make no sense for the third match of a team to use the lags of the last 2 weeks, 6 months or 5 years, since we only have records of two matches of that team, so the three lags would have the same value when they represent totally different time intervals. The same is true using the last n games. Therefore, a minimum number of historical matches has been set for each lag.

Table 3.8 shows an example. Here it is shown how for the first two matches of the season we do not have data for the last 15 and 60 days, because a minimum of two matches are required to create the value of these lags. The same applies to the 15-day lag for matchdays 5 and 6, as we do not have enough data for the last two weeks due to the international break. However, the 365-day lag does contain value for these matchdays because, although the minimum is 20 matches in the last year, this is something that is satisfied as long as the team has participated in the league the previous season. If a team has just been promoted that same season, we would not have data for this field (365 days) until the 21st matchday.

Regarding the variables of the history between the two teams (*Derby*), all the matches have historical values, but it could happen that a team makes its debut in the category or

has been in the competition for a few seasons (at least since we have data) and the last n games between the two teams are not registered.

Date	Round	Home team	Away Team	Result	Points			Derby			
					15D	60D	365D	1	3	5	10
12/08/2007	1	Arsenal	Fulham	2-1			1.79	3.0	3.0	3.0	3.0
19/08/2007	2	Blackburn	Arsenal	1-1			1.82	3.0	1.0	1.8	1.5
25/08/2007	3	Arsenal	Man City	1-0	2.0	2.0	1.82	3.0	3.0	3.0	2.71
02/09/2007	4	Arsenal	Portsmouth	3-1	2.0	2.33	1.89	1.0	2.0	2.0	2.0
15/09/2007	5	Tottenham	Arsenal	1-3		2.5	1.95	1.0	1.0	1.0	1.4
22/09/2007	6	Arsenal	Derby	5-0		2.6	1.95	3.0	2.33	1.6	1.83

Table 3.8: Example of lags of points gained for Arsenal FC on the season 2007-2008.

A description follows of the different features in the different dataset versions and of the way in which they have been obtained:

- **Side_ t_S ³**: represents how much the team S has played as home or away team in the matches within the last t time. To calculate this, an auxiliary variable was created in the original dataset in which 0 was assigned to the home team and 1 to the away team. The value of this new attribute is the average of the auxiliary attribute of each team during this time period t . Therefore, 0 as the value of this attribute means that all the matches have been played at home and 0.5 means that half have been played at home and half away. The attribute tends to 0.5 as the time lag gets bigger.
- **Scored_ t_S** : sum of scored goals by the team within the t time lag.
- **Received_ t_S** : sum of received goals by the team within the t time lag.
- **Points_ t_S** : mean points gained by the team within the t time lag.
- **Shots_ t_S** : sum of shots by the team within the t time lag.
- **Target_ t_S** : sum of target shots by the team within the t time lag.
- **Corners_ t_S** : sum of corners taken by the team within the t time lag.
- **Faults_ t_S** : sum of fouls committed by the team within the t time lag.
- **YellowCards_ t_S** : sum of yellow cards received by the team within the t time lag.
- **RedCards_ t_S** : sum of red cards received by the team within the t time lag.
- **Derby_feature_ n_S** : mean of the points, goals and *Side* features of both teams in the last n matches against each other. This variable is meant to provide the model with historical information on the performance of both teams in the last 3, 5 and 10 matches between them.

Finally, it is important to highlight the reasons why the different versions change in number of matches with respect to the original dataset. The reason why matches are lost when training the model is that all records must have the same dimensionality; however, there are many matches for which there are no values available for some variables (lags) and, therefore, those matches that do not contain all the values are discarded.

³ t represent the lag time. As it is shown in Table 3.7, depending on the dataset version, it can be either last n games or the last d days. S represents the home and away team.

First of all, in order to create the different lags, we lose the entire 1993-1994 and 1994-1995 seasons, as for many lags the minimum number of matches required is not reached. In the following seasons we also miss quite a few matches, especially for teams that are new to the competition. Finally, there are matches that are missed during all seasons, such as the first matchdays of the season or matches after the international football breaks.

In order to minimise the damage of dropping many matches during the experiments, different dimensionalities and different values of minimum matches required to create a lag have been tested to find a balance between number of records and performance of the model. Furthermore, the experiments have been implemented in such a way that it is possible to pass as a parameter the features we want the training data to have. Thus, we can eliminate features (such as the 15 days or the derby of the last 10 matches) that make us discard many matches. This functionality is used to find that balance between the number of features, the information provided and the number of records to be trained, and it can also be useful to eliminate certain variables that we detect empirically that do not add value to the model. This, however, will be discussed in detail further on.

3.3 Data analytics

This section consists of the analysis of the datasets at a statistical level, first in order to know and understand how data is distributed, what patterns it follows and what outliers has, and then in order to be able to make better decisions when creating the datasets, selecting features and training the models.

3.3.1. Prior dataset creation analysis

The first analysis was carried out before creating the datasets with the aim of finding patterns and anomalies in the data that could give us hints about which features may influence the outcome of a match and which ones may not influence it at all.

The first thing to know is how balanced the dataset is. Table 3.9 shows that the dataset is unbalanced since almost half of the data, 45.7%, were home wins, while 29.5% ended up in away wins and only 24.8% in draws.

Draw	Home win	Away win
24.8%	45.7%	29.5%

Table 3.9: Distribution of the match results.

Local victories have always been predominant (Figure 3.1), surpassing in some seasons and leagues more than half the number of matches, as in the Bundesliga 1996-1997 and LaLiga 2010-2011. Nevertheless, some surprising conclusions can be taken from this graph. One of them is the significant general increase in the number of matches that end in away wins. While 25 years ago, away wins were around 20%, in the last few seasons it has consistently exceeded 30%, and even in the 2020-21 season in the Premier League and Ligue 1 it has exceeded and equalled the number of home wins respectively. It can be seen how 20 years ago the away wins line was below the number of draws and for the last 10 to 15 years the away wins have overtaken the number of draws.

In contrast, home wins and draws have fluctuated over time, though with a slight decreasing trend over time. It is possible that the sudden drop in the 2019-20 and 2020-21

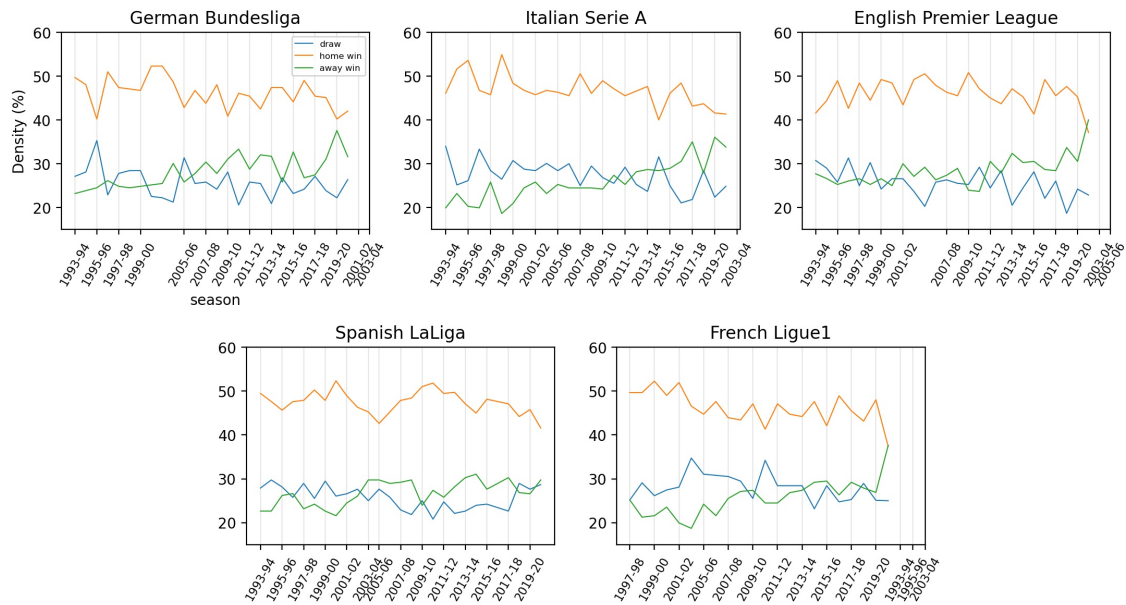


Figure 3.1: Distribution of match results since 1993 to 2021 in the five leagues.

seasons in home wins and the rise in away wins is related to the capacity ban at football stadiums during the Covid-19 pandemic. In the French league graph, for example, we would see that in 2019-20 there is no significant change compared to 2018-19 as this competition was suspended when the pandemic and lockdown was declared. However, the following year, in 2020-21, when the league was played without a capacity ban, away wins spiked and home wins dropped.

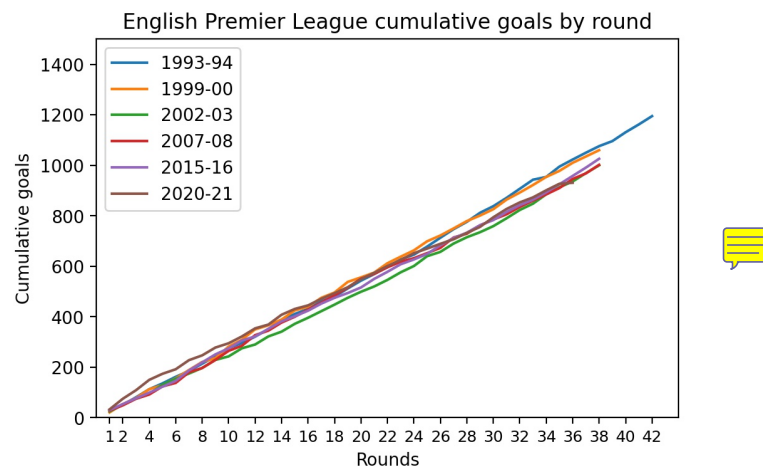


Figure 3.2: Cumulative goals distribution in different English Premier League seasons.

As we have seen, the distribution of results is not uniform over time or in each competition. As well as the results, other statistics have their own evolution over time. Figure 3.2 shows how over the last 30 years the distribution of goals scored per matchday in the Premier League remains very similar. While it is true that 25 years ago it seems that more goals were scored in the final rounds of the season, it can be assumed that goals scored follow a very similar distribution over time, so using goals scored in 1995 matches to predict today's matches should not be a mistake.

Figure 3.3 shows the distribution of goals scored per match for the five leagues in the 2017-2018 season. In the case that goals are considered to be events that can happen with

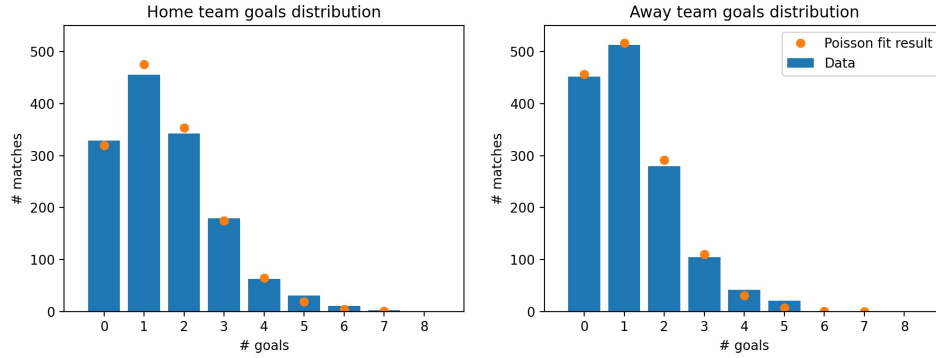


Figure 3.3: Goal distribution with fitted Poisson distribution

a given probability during the 90 minutes, and it is assumed that matches are independent of each other and that goals $\Rightarrow$ match are independent of each other, it could be said that they follow a Poisson distribution. As shown in Figure 3.3 both distributions are very close to Poisson. Home goals follow a distribution with $\lambda = 1.485$ and away goals with $\lambda = 1.13$. Another conclusion that can be made from the two distributions is that the home team is slightly more likely to score than the away team. From the Poisson PDF $\Rightarrow$ it is calculated that the away team has nearly a 70 percent probability of scoring none or one goal, while the home team has only a 56% percent probability of scoring none or one goal..

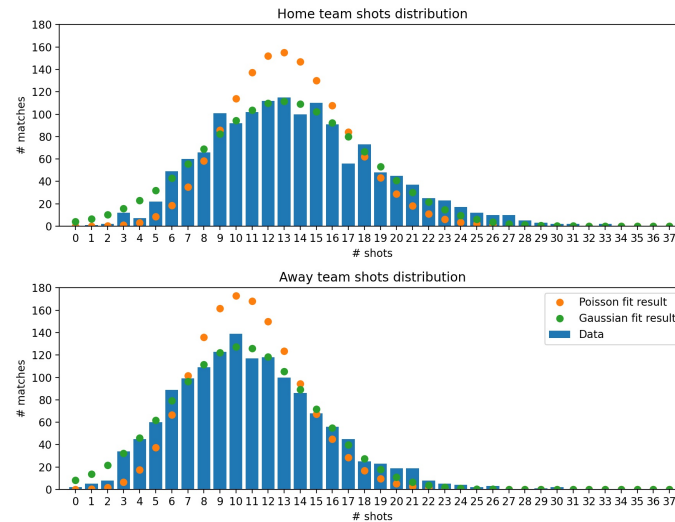


Figure 3.4: Shots distribution for season 2017-2018 with fitted Poisson and Normal distribution.

It is also interesting to see the distribution of the number of shots taken in a match. Figure 3.4 shows that this feature fits better with a Normal distribution, with parameters $\mu=13.27, \sigma=5.27$ and $\mathcal{N}(\mu=10.86, \sigma=4.55)$ respectively. The conclusion that can be taken from this is that home teams are more likely to take more shots than away teams.

The distribution of shots taken by the winning and losing teams can give a (fairly intuitive) idea of whether there is a clear relationship between the number of shots taken and the outcome of a match. Excluding matches that ended in a draw, these are the distributions.

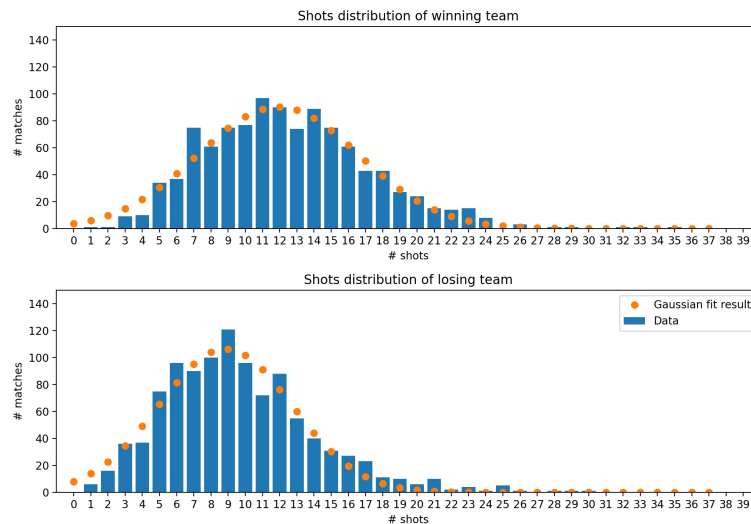


Figure 3.5: Shots distribution for season 2017-2018 of winning and losing teams with fitted Normal distribution

Finally, in terms of shots, it is worth comparing the distribution of the difference in shots between the winning and losing team. There are two scenarios here, when the team with the most shots wins and when the opposite happens, the team with the fewer shots wins (Figure 3.6).

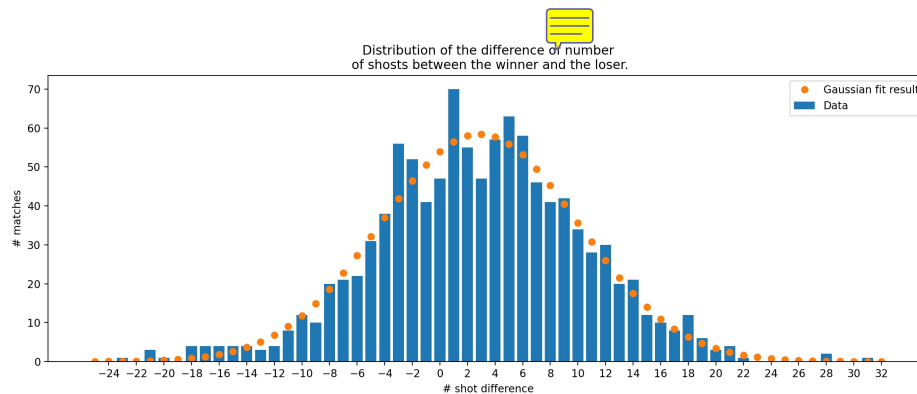


Figure 3.6: Shot distributions between winner and loser at season 2017-2018.

It can be observed that there are slightly more matches in which the winning team takes more shots, but the difference is not very large, which means that there are many matches that are very evenly matched on the pitch and that the result ends up going one way or the other by little details. It is not uncommon for a team to take more than 10 shots and end up winning; however, there are matches where the opposite happens. These cases are especially difficult to predict for a predictive model that takes as input a time series of events [Nyquist and Pettersson, 2017]. The team that takes the fewest shots ends up winning in 32% of games.

Another fundamental tactical aspect of football is passing. Figure 3.7 shows how winning team tends to make more passes during the match. In almost 25% of the matches the winning team makes between approximately 550 and 850 passes, while the losing team makes between 450 and 650 passes. It is true that the median and the first quartile (Q1) of each distribution are practically the same, but the third quartile (Q3) and the upper range of typical values vary a lot between the two distributions. The conclusion that

can be drawn is that if a team makes more than 600-700 passes it is very unlikely to lose the match.

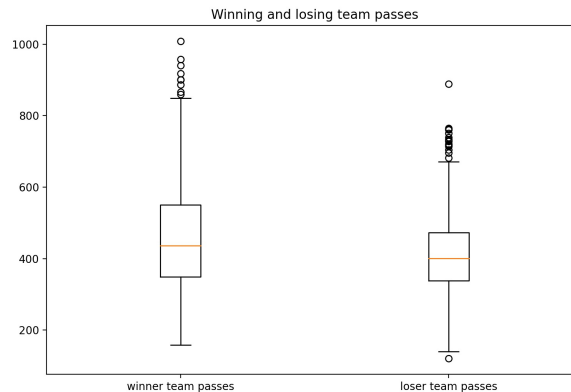


Figure 3.7: Winning and losing team passes box plot.

rank.	team	pos.	acc.
1	Manchester City	1°	0.9
2	PSG	1°	0.89
3	Nice	8°	0.89
4	FC Barcelona	1°	0.88
5	Bayern Munchen	1°	0.88
...			
94	Leganés	17°	0.78
95	Stoke City	14°	0.77
96	Augsburg	12°	0.77
97	Crotone	18°	0.76
98	Getafe	8°	0.73

Table 3.10: Top 5 teams with most pass accuracy and its position in the 2017-2018 league table and the top 5 teams with the less accuracy at the pass.

But it is not only the quantity of passes that is important, but also their quality. Table 3.10 shows how in the top five teams with the best passing accuracy there are 4 league champions. In the five teams with the least passing accuracy it is more difficult to draw conclusions because although they are teams in the lower-middle part of the table, there is only one team among them that was relegated, Crotone. It is curious to see how Nice and Getafe both finished eighth in their respective leagues. The fact that one is the third with the most goals and the other is the least (with a big difference from the rest) is due to the style of play of both teams, while Nice bets more on possession and control of the game (like the styles of play of Manchester City or Barcelona), Getafe has a more defensive, aggressive and counter-attacking style of play. We can therefore see how a team's style of play can be inferred from some of their statistics.

However, football is all about goals, and although, as we have already seen, passing is important to win, shooting is even more essential to win matches. Table 3.11 shows the teams with the most and least goals scored and the goals they have scored during the season. It can be seen that the top scoring teams have very good shots, which means that they have very good strikers. It is remarkable that Hertha Berlin appears in second

rank.	team	goals	acc.
1	FC Barcelona	99	0.48
2	Hertha Berlin	43	0.44
3	PSG	108	0.42
4	Real Betis	60	0.42
5	Manchester City	106	0.41
...			
94	Sassuolo	29	0.3
95	Amiens	37	0.29
96	Nantes	36	0.28
97	Hellas Verona	30	0.28
98	Caen	27	0.27

Table 3.11: Top 5 teams with more and less shot accuracy and the total amount of goals they scored in 2017-2018.

position. The teams with the least shot accuracy are also the teams that score the least goals.

The style of play of an opponent is one of the characteristics that coaches consider most often when studying their opponents. The strategies, tactics and line-ups that form a team's style of play are not static over time, but vary over time, depending on the coach, the players, the opponent and the context of the match. A team will not apply the same strategy on the first match day, when there is nothing at stake, as on the last match day when they are playing for avoiding relegation; in the same way, a team in the bottom half of the table will not apply the same strategy against a similar team as against a team in the top part of the table.

Therefore, styles of play interact with each other, and influence the final outcome of a match. In Figure 3.9 can be seen how opponents who in the last 15 years have beaten Barcelona - a team that has a very dominant, attacking possession style of play - use different playing strategies. The teams at the top of the table rarely have fewer than 10 shots for the whole match, suggesting a more attacking approach due to the equal level of the two teams. Very few teams have beaten Barcelona by dominating the game offensively (with more than 15 shots), and none of these teams were ranked lower than 14th in the table. This may suggest that teams prefer to use a defensive and counter-attacking style of play against Barcelona.

Another interesting aspect of the style of play to analyse is the fouls committed by the team that beat Barcelona, and thus the aggressiveness of the team. Figure 3.9 shows, through the smaller points, how many teams that normally do not have a particularly aggressive style of play, make a lot of fouls against Barcelona, surely due to the defensive strategy they follow. This is the case of Atlético de Madrid in the 2020-2021 season or Celta in the 2014-2015 season. Both were teams that during the season did not commit many fouls; however, they used a defensive and aggressive strategy that gave them a good result against Barcelona. Almost every team that beat Barcelona committed more fouls in that match than the season average, suggesting that employing an aggressive and very intense style of play increases the chances of winning against Barcelona.

Moreover, the style of play is not constant over time. Over the course of a season, despite different strategies are used against different opponents, there is always a certain style of play that predominates. Figure 3.8 shows the evolution over time of goals scored

and conceded, as well as yellow cards received by three teams from different leagues and backgrounds: Atalanta (Italy), Getafe (Spain) and Schalke 04 (Germany).

It can be seen in the graph on the left (Figure 3.8) the trend towards offensive style of play at Atalanta Bergamo since 2016, just since Gasperini's arrival as coach. In 10 years Atalanta went from 2nd Division to the top positions of Serie A by 2020 while being the highest scoring team and competing in the UEFA Champions League playing one of the most attacking and entertaining football in the old continent. It is interesting to see how the trend towards ultra-offensive, match-dominating football meant that the team needed to commit fewer fouls, going from an average of 3 fouls per game in 2012 to 1.5 in 2020.

A completely opposite success story, but one that also shows how a coach can radically transform the way a team plays and performs, is that of Getafe. Between 2017 and 2019 they experienced a big transformation in their style of play and managed to balance an aggressive game with a great defensive strength that led them back to competing in European competitions. This change coincided with the arrival on the bench of Bordalás (characterised by this style of play) during the team's stay in the 2nd Division in 2016-2017 and until his departure in 2020.

Finally, a case of a negative trend in performance is shown in the case of German Schalke 04. During the first decade of the 21st century and until 2015, it lived a golden era being one of the best German clubs of the new century, and although not winning the league, it was the runner-up in the league in 2005, 2007 and 2010, cup winner in 2001, 2002 and 2011, and participated 6 times in the UEFA Champions League, reaching the semi-finals in 2011. However, since 2018, they fell into a spiral of bad transfer policies and poor results that culminated in their relegation as bottom club and making only 13 points in the league, a negative record in the Bundesliga. You can see how the few last bad years have changed the way the team plays, and surely the loss of quality in the squad has led them to play a less offensive and aggressive style of football.

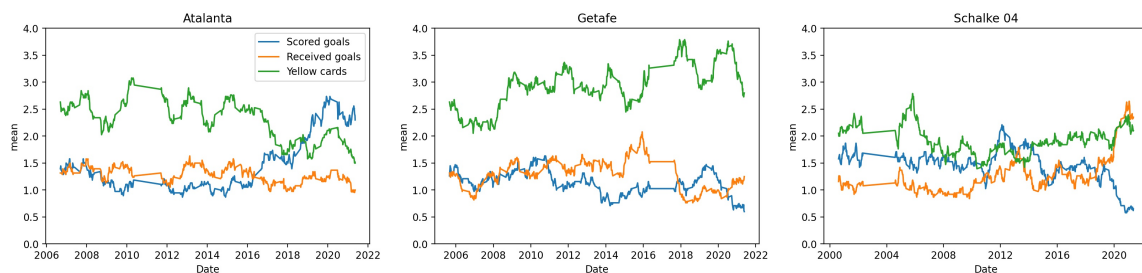


Figure 3.8: Evolution of the scored and received goals and yellow cards for PSG, Getafe and Schalke 04.

To conclude the analysis of the data, the relationship between the betting odds of BET365 and the outcome of a match is shown. The aim of this is to see whether the odds hide a direct relationship with the probability of a team winning, and the degree of this relationship. In reality, the odds represent the inverse of the probability assigned by the bookmaker to an outcome. Therefore, the Figure 3.10 shows the inverse of the odds.

Bookmakers predict accurately most of the matches in which there is a high probability that one or the other team will win. They predict almost all the matches from a probability of 0.7. Line with function $f(x) = 1 - x$ represents probability 1. When compared to

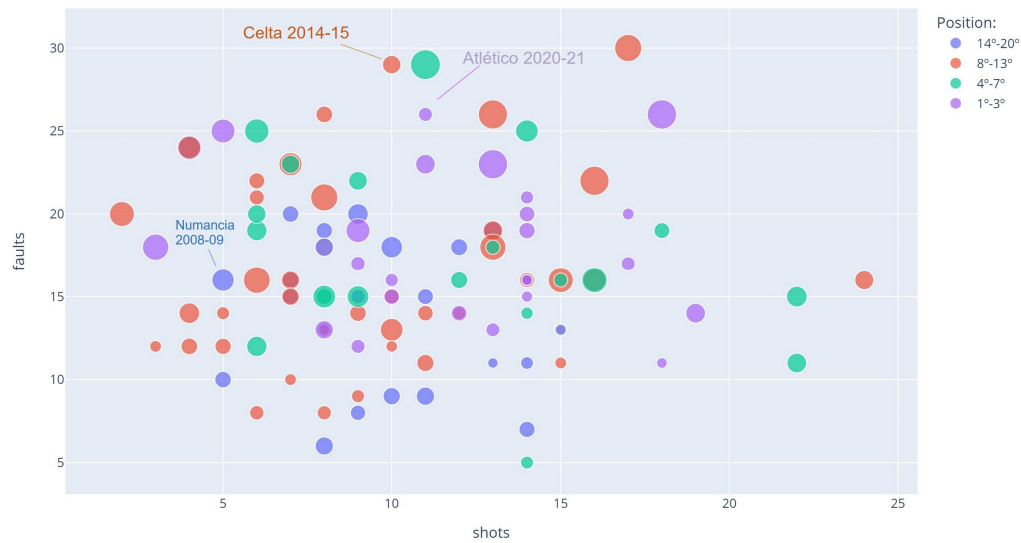


Figure 3.9: Scatter plot distribution of shots and fouls. The size of the point shows the mean of the fouls of that team in the match season, and the colors shows the position of the team in the table that season.

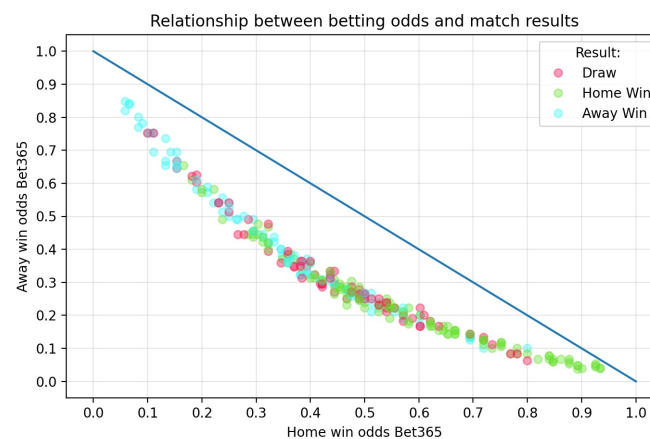


Figure 3.10: Bet365 probabilities based on betting odds.

the points on the probability curve of *BET365*, it is found that the distance between the curve and the points is the probability implicitly assigned by the bookmaker to the draw. It should be noted that bookmakers always assign a higher probability to the victory of one of the teams rather than a draw, making many mistakes in the middle of the graph where the matches are more evenly-matched.



3.3.2. New dataset analysis

In this subsection, some of the features generated in both datasets are statistically analysed and compared with the original features.

First of all, if comparing the distributions that follow the shot features generated in the first dataset, *shots_11H* y *shots_11A*, with the shot distributions in the source dataset (Figure 3.4) it can be seen that both distributions have equal shapes. This is because these

generated attributes are the sum of the shots of a team's line-up players in the last 4 home and away matches (Figure 3.11), so it is logical that they follow similar distributions.

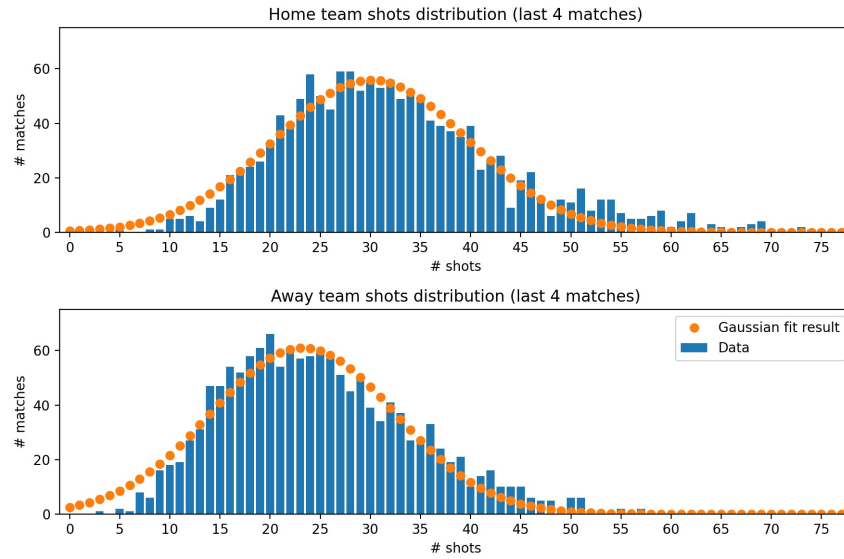


Figure 3.11: *shots_11H* y *shots_11A* distribution and fitted with Gaussian distribution.

Two completely new features generated in the new dataset are *mins4_H* and *mins4_A*. In Figure 3.12 it can be seen that they follow a distribution that can be fairly close to a Normal distribution, especially in the "slopes" of the PDF of the distribution, since in the area of the mean (the "peak") it fits worse and the distribution underestimates those values. The Figure 3.13 shows the relationship between both attributes and the match result, giving as a conclusion that there seems not to be an evident relationship between both variables, so they appear to be independent. Note that the purpose of these attributes is to indicate to the model whether a team plays with its usual eleven or with new players, a factor that is likely to influence the outcome of a match.

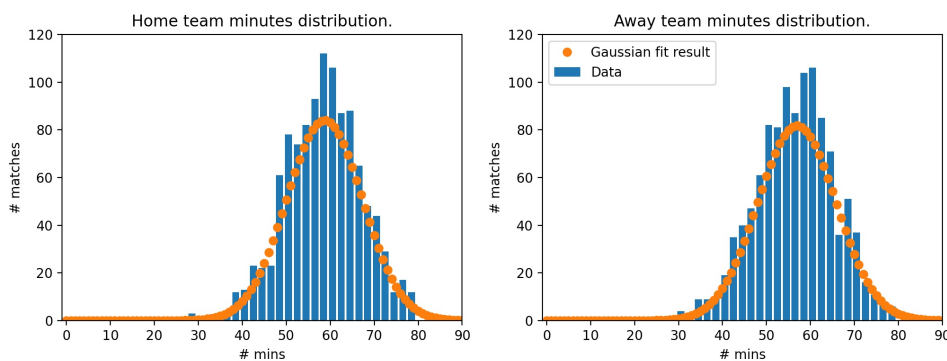


Figure 3.12: *mins4_H* y *mins4_A* distribution and fitted with Gaussian distribution.

Table 3.12 shows an indicator created to measure the attacking strength and defensive weakness of each team. For the model we will use two separate attributes for each team, *attack H* or *A* and *defence H* or *A* which will use the goals scored and conceded respectively to indicate this. However, in this table we show the ranking of the indicator (unifying attack and defence) considering all the matches of the season, in order to show how this indicator shows very well the level and the results of the teams.

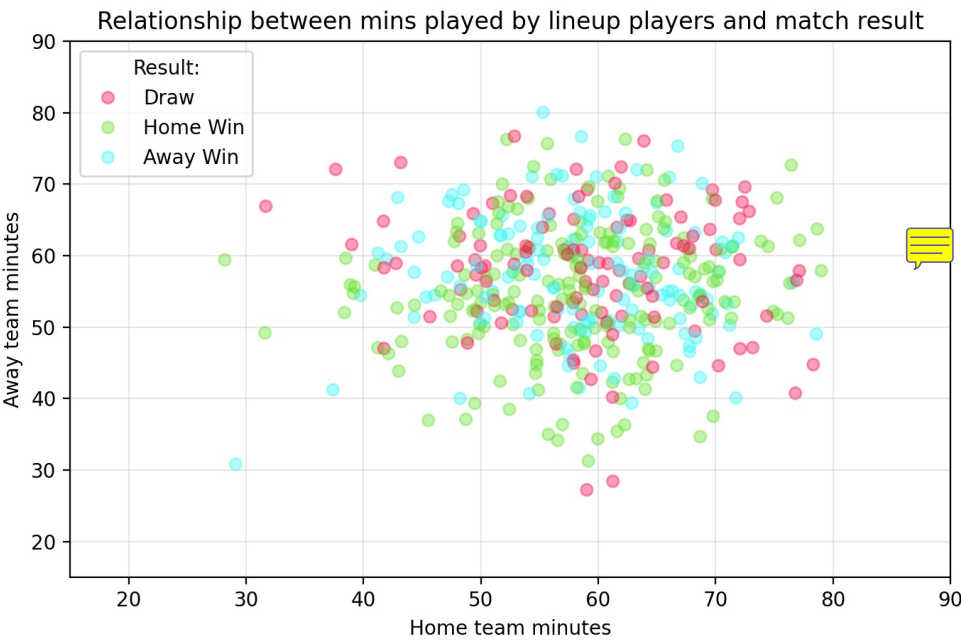


Figure 3.13: Relationship between *mins4* and *mins4_A* with match results.

rank.	team	pos.	att.-def.
1	PSG	1°	2.02
2	Juventus	1°	1.99
3	Bayern Munchen	1°	1.98
4	Manchester City	1°	1.94
5	FC Barcelona	1°	1.83
...			
94	Huddersfield Town	16°	0.69
95	Málaga	20°	0.68
96	Hellas Verona	19°	0.66
97	Caen	16°	0.65
98	UD Las Palmas	19°	0.57

Table 3.12: Top 5 teams with more and less attack-defense indicator in season 2017-2018.

On the other hand, the variables based on "lags" created in the Historical training dataset are intended to show the model the short, medium and long term trends of the teams. Short lags will show the model the most recent form of the team, while longer lags will show what the historical status of the team is, thus having complete information over time. Figure 3.14 shows how the lags of Atalanta, Getafe and Schalke 04 evolve over the seasons, which can be compared with the time series in Figure 3.8. It can be seen from the confidence interval shown (shaded area) that the shorter lags have much more variability, while the longer lags are much more constant and their line has a much smaller and more regular slope. The model will have to learn which is more important for deciding the outcome of a match, the recent streak or the long-term streak.

Table 3.13 shows three games in which Chelsea, one of the best English teams in the last 20 years, came from a poor form and faced teams that were historically worse but were on a very good run of form. The first two matches are from the 2015-2016 season



and Chelsea had been league champions the previous year, but this season they suffered a setback and finished 10th. In contrast, Leicester City, who had only been in the Premier League for one year, achieved one of the greatest triumphs in football history by winning the league that year. Therefore, both indicators of the streaks of the two teams are important in predicting an outcome. This example shows how the model will receive information about both Leicester's good year and Chelsea's bad year, as well as knowing that Chelsea and Leicester's level are likely to be temporary. The model will know this because of the long term level of the two teams (lag of 730 days).

In Table 3.13 two matches are shown which are won by the teams that were less favourite (less points in the long term) but which arrived with a better short term run; however an example is also shown of a match, Chelsea vs Burnley, which is won by the favourite team because that is why they are the favourites. Therefore, the model should learn from these and the other variables whether the short-term or long-term streak is more decisive for a match.

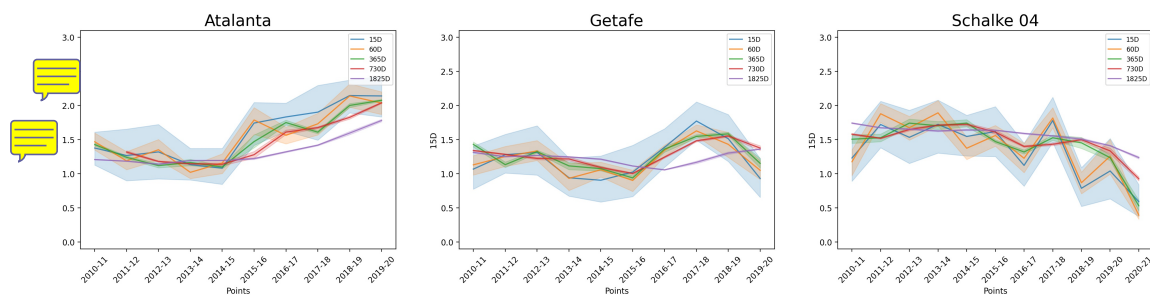


Figure 3.14: Time Series of the different lags in historical dataset for every season. The dark line represents the mean of a lag in a season, and the shaded area represents the 95% Confidence Interval.

Date	Home team	Away team	Result	points lag 60D home	points lag 60D away	points lag 730D home	points lag 730D away
2015-10-24	West Ham	Chelsea	2-1	2.33	1.16	1.25	2.12
2015-12-14	Leicester	Chelsea	2-1	2.43	1	1.38	2.03
2018-04-19	Burnley	Chelsea	1-2	2.67	1.17	1.3	2.09

Table 3.13: Example of lags and results for three Chelsea matches where Chelsea arrived with in a bad shape against another a priori smaller team that arrived in a very good shape.

In addition to the lags, variables have been created based on the history of the match concerned (Table 3.6). There is a relationship between the outcome of a match between two teams and their previous matches against each other (Figure 3.15). When the home team has dominated the last meetings (more than 2.5 points on average) it is very likely to win the next one. If the last meetings have been closer (between 1 and 2 points on average) the relationship with the outcome of the next match is weaker and any outcome is likely, although it still seems that the home advantage has a considerable influence and the home team is (a priori) more likely to win.

Also the home advantage has an influence on the fact that it is easier to break a winning streak of the away team than one of the home team. It can be seen in the last 3 games how it is easier for the home team to win despite the away team having dominated the last few games than vice versa.

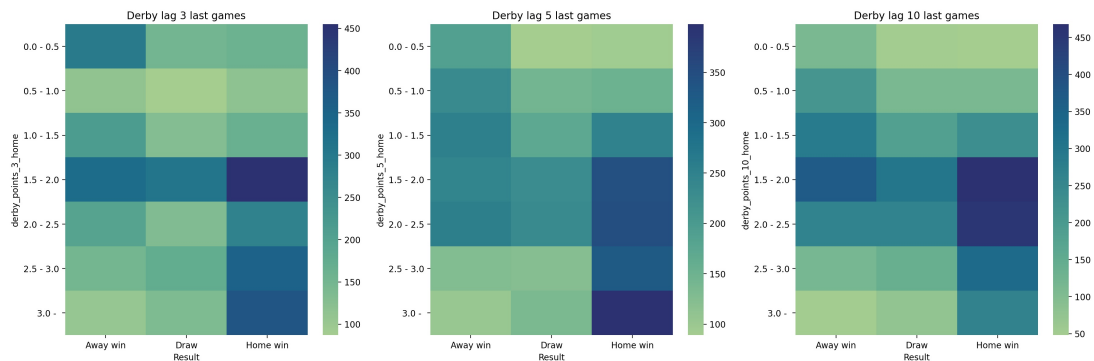


Figure 3.15: Relationship between the final result of the match (x axis) and the distribution of home points in the last n games between the two teams (y axis) in top 5 european leagues from 2018-19.

3.4 Dimensionality reduction

As part of the feature engineering and feature analysis, it is explained below how the dimensionality reduction process of the data has been performed. Dimensionality reduction is responsible for identifying, selecting and removing features that do not contribute to or even degrade the performance of the model. Two main types of dimensionality reduction techniques are shown below. These techniques will be applied to the 20-feature dataset created from the Wyscout based dataset data.

3.4.1. Feature selection

There are many techniques for feature selection but three techniques will be used, a univariate analysis technique, *Variance Threshold*, and three multivariate analysis techniques, *ANOVA* and *Pearson correlation*. Before applying the feature selection techniques, data is normalised so that it is on the same scale. The normalisation has been performed with the *normalize* function of the *preprocessing* package of the Machine Learning *Scikit-Learn* library.

Variance Threshold The first and simplest feature selection method used is Variance Threshold. Variance Threshold is often used as a baseline to compare with more complex methods, and uses a threshold to drop all those variables whose variance does not surpass it. This reduces the dimensional data space D to a space of d dimensionality, with $D > d$. The variance measures the spread of the values of a variable, it measures how far a value is from the mean of the variable. This technique is used with the approach of training the model with the variables that have more variance, since it is likely that the fact that the data are more separated in the space of dimension d increases the separability of the data with respect to a space of the same dimensions but with variables with less variance.

During the training of our model, different thresholds will be used to determine which threshold selects, according to the variance, the optimal number of variables.

Some conclusions can be drawn from Figures 3.16 and 3.17. First, in Figure 3.16, pass success rate may not be a feature that can contribute much to our model because there is no significant variance in it. In Table 3.10 it was already apparent that this was the case due to the small range of the data, between 0.73 and 0.9. A similar thing happens with

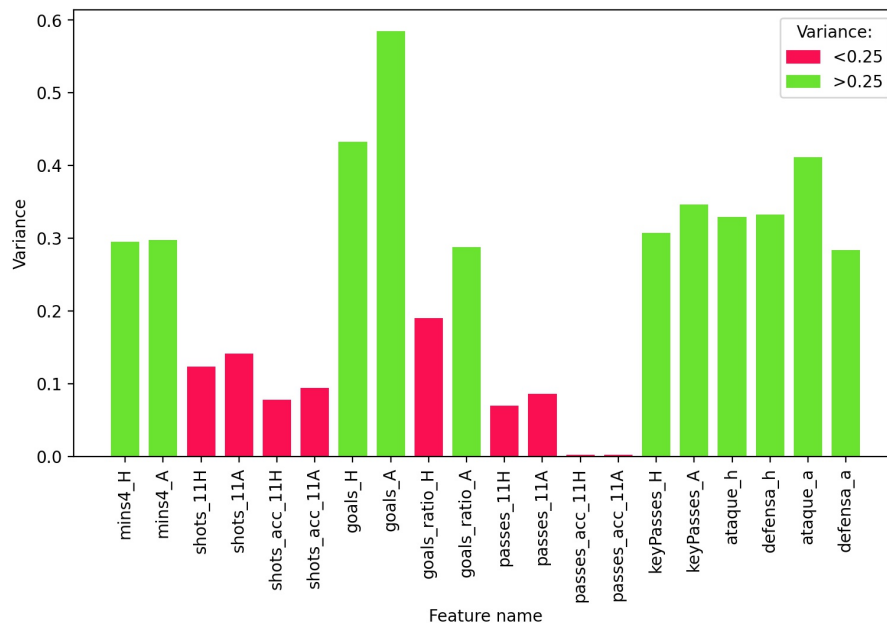


Figure 3.16: Normalized Wyscout based dataset feature variances

shots, passes and shots on target, although in this case there is some variance and we will have to see the results of other methods to draw conclusions. On the other hand, goals are the variables with the highest variance, and therefore it is possible that it is the variable that helps our model most in order to know which teams are in the best shape.

Figure 3.16 shows in green the features that are selected with a threshold of 0.25, a total of 11 features. It mostly selects in pairs, e.g. from goals it selects both home and away goals, as the distributions that follow are quite similar as we have seen above (Figure 3.11).

Regarding the variance of the Historical features, it can be seen in Figure 3.17 how the shortest lags are the ones with higher variance, as it has been explained in 3.14. It is also interesting to highlight how points, goals and red cards are the features with more variance across all the lags, while features like fouls or corners are more uniform and similar in every match.

ANOVA Analysis of Variance is a parametric statistical hypothesis test for determining whether the means from two or more samples of data come from the same distribution or not. ANOVA shows the relationship between a categorical variable and a continuous variable, i.e. whether one depends on the other. More specifically, it shows how the continuous variables change with respect to the categorical variable. In this case, assuming independence of the samples, One-Way ANOVA will be used to find the relationships between the characteristics (continuous variables) and the class or outcome (a single categorical variable).



Figure 3.18 shows the p-values resulting from applying ANOVA to our data. Values close to 1 indicate a dependence between the variable and the outcome of the match (categorical feature), while values close to zero show independence. These independent variables will decrease the performance of our model as they do not provide information to classify the match.

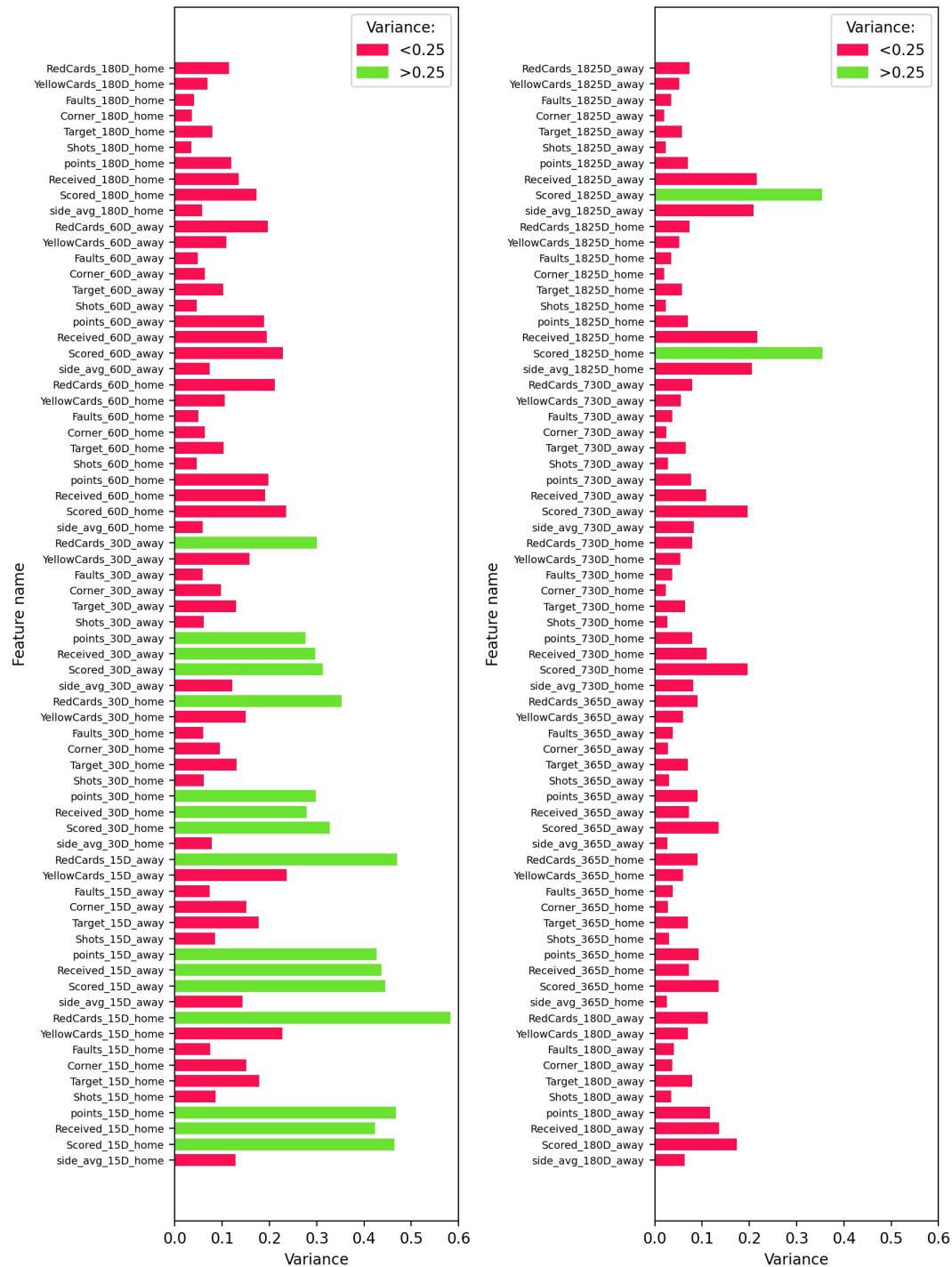


Figure 3.17: Normalized Historical dataset feature variances

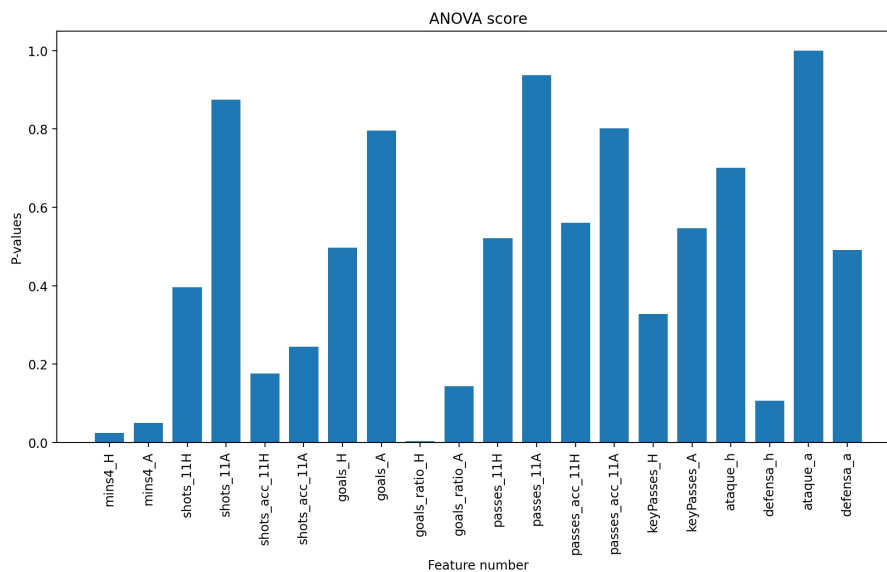


Figure 3.18: ANOVA F-value score showing the dependence between continuous variables and categorical target variable.

The ANOVA analysis shows that, surprisingly, the characteristics of the visiting teams vary more depending on the outcome of the match. No explanation for this has been found so far. Another surprising observation is the dependence between the result and the passing success. It was seen in the Variance Threshold (Figure 3.16) how these variables had a variance close to zero. There also exists a strong dependence of the results on shots, goals, total passes, key passes and attack indicator in the last four games, possibly due to the strong dependence between key passes, shots, goals and the attack indicator. It is also remarkable that there is very little dependence on the minutes played by the players in the last four matches, as already seen in the dataset analysis (Figure 3.13), and on the ratio of goals scored in the last four matches by the players in the starting eleven.

Figure 3.19 shows the ANOVA scores for the Historical dataset, and the top 50 features with more correlation with the target are coloured in green. It can be seen immediately that the features selected are completely different from the ones in Variance Threshold (Figure 3.17). This may be because the favourite team - the one with a better long term streak - usually wins the matches. However, this ANOVA score is not a reason to immediately remove the non-selected features from the dataset (red ones in the figure), as these features can provide meaningful information to the model in some cases, as it has been analysed previously. It is worth mentioning that there are features that may be strongly correlated between them - such as goals, shots or corners - providing tautological information to the model. But this will be discussed in the following subsection.

For selecting the features most dependent on the target feature we have used `SelectKBest` class from the *feature selection* package of the *Scikit-Learn* library, and we have trained ANOVA with our data. The number of variables to be selected is indicated by the parameter *k* which is passed as a parameter to the initialisation of the `SelectKBest` object. This parameter *k* will be trained with our model to identify the optimal number of variables.

Pearson Correlation The Pearson correlation coefficient measures the linear relationship between two continuous variables. This one varies between -1 and +1, with 0 implying no correlation. Correlations of -1 or +1 imply an exact relationship. Positive values

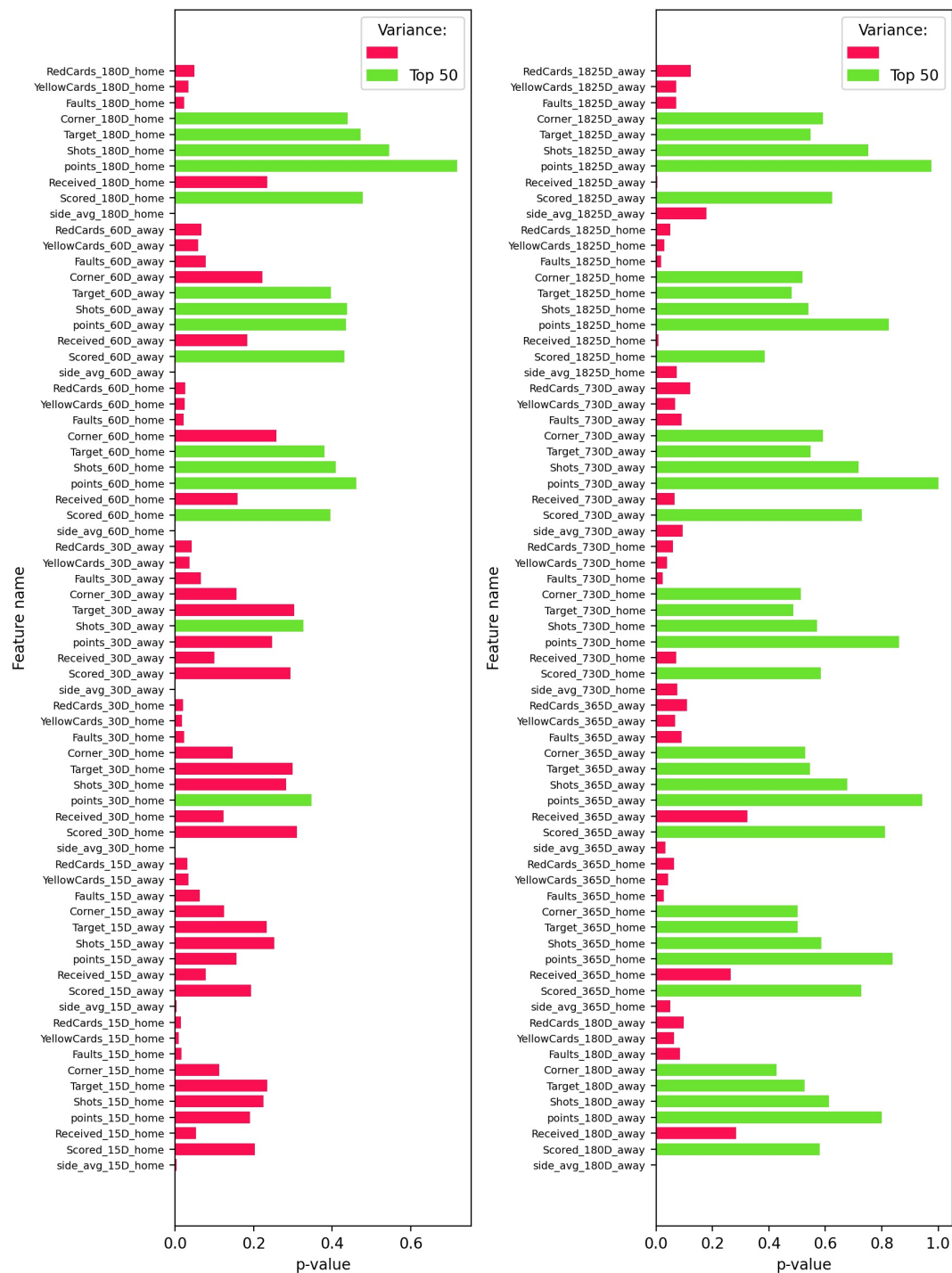


Figure 3.19: ANOVA F-value score showing the dependence between continuous variables and categorical target variable in 3.2.2

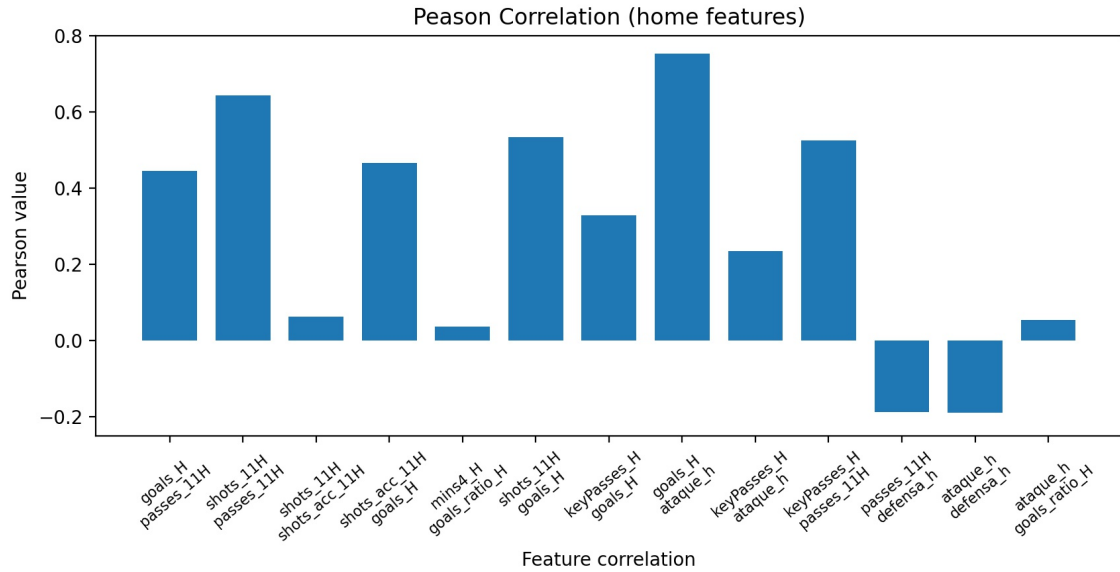


Figure 3.20: Pearson correlation between some home team features of the last four matches as home team.

imply that as x increases, so does y , and they are positive correlations. Negative values imply that as x increases, y decreases, and they are negative correlations. The Pearson (ρ) is essentially a normalized measurement of the covariance, it is the ratio between covariance of two variables ($cov(X, Y)$) and the product of their standard deviations (σ).

$$\rho_{X,Y} = \frac{cov(X, Y)}{\sigma_X \sigma_Y} \quad (3.1)$$

All attributes have been compared with each other to find dependencies between two characteristics to see if it is possible to discard variables due to a high dependency on another. These are the strongest and weakest dependencies (Table 3.14):

Highest dependency			
rank	var 1	var 2	p-value
1	mins4_H	mins4_A	0.9
2	goals_A	ataque_A	0.77
3	goals_H	ataque_H	0.75
4	passes_11A	passes_acc_11A	0.68
5	passes_11H	passes_acc_11H	0.65

Lowest dependency			
rank	var 1	var 2	p-value
1	goals_H	passes_11A	1.0e−5
2	goals_A	ataque_H	1.97e−4
3	shots_acc_11A	goals_ratio_H	1.1e−3
4	shots_acc_11H	goals_A	1.2e−3
5	shots_acc_11A	passes_11H	3.3e−3

Table 3.14: Pearson correlations for Wyscout dataset.

It can be seen in Table 3.14 the strong dependence between the variables of goals and attack, as had been assumed in the ANOVA analysis (Figure 3.18). Figure 3.20 compares the p-values of several local team variables. The strong dependence between passes, key passes, shots and goals can also be seen. The smooth negative dependence between passes and the defence indicator is also observed, meaning that when a team makes a lot of passes it is likely to concede few goals. Pearson also highlights the independence of some variables as we can see both in the figure and in the table, for example independence between the number of shots and goal scoring, the minutes played by the starters and the ratio of goals scored by the team in the last four games. In the table of lower dependencies (Table 3.14) you can see, as you might have guessed, that the most independent variables are those of one team with respect to the variables of the other team. This analysis applies both for the Wyscout and the Historical dataset.

From this analysis some interdependent variables can be discarded, also taking into account the Variance Threshold and ANOVA analyses, and assuming that this will improve the performance of the model. Therefore, for the initial dataset (3.2.1), if we made a manual selection of variables from the analysed, we would take these twelve variables:

shots_11H	shots_11A	shots_acc_11H	shots_acc_11A	goals	goals_A
passes_11H	passes_11A	ataque_H	defensa_H	ataque_A	defensa_A

Table 3.15: Manually-selected features based on intuition and feature selection analysis

The performance of the model with these selected variables will be compared with the performance of the variables selected with the other dimensionality reduction methods.

3.4.2. Principal Component Analysis (PCA)

This data dimensionality reduction technique is a different approach to feature selection techniques. It is a supervised learning technique in which the dimensionality of samples is reduced by calculating the first n principal components, which transforms the original data to a smaller dimensional space while keeping as much information as possible contained in the original data.

The principal components of a set of samples or points in a d -dimensional space are a set of d unit vectors, where the i -th vector indicates the direction of the line on which, if the points are projected, they maintain the maximum possible variance of the projected points and in turn the vector is orthogonal to the previous $i - 1$ unit vectors. Therefore, the problem faced by PCA is the mapping of the data into a lower-dimensional space, maximising the variance of the projected data in this low-dimensional space. In other words, the aim is to reduce the number of features with the minimum possible loss of information. It can also reduce existing noise in the original data.

PCA is usually explained and implemented with the calculation of eigenvectors (matrix V , Eq. 3.3) and eigenvalues (diagonal matrix, L , Eq. 3.3) of the data covariance matrix (matrix C , Eq. 3.3) which we pretended to transform. The first d eigenvectors correspond to the n -th first principal components to which the data are projected (matrix X) obtaining then the transformed data into a d -dimensional space (Eq. 3.4). The percentage of variance retained in the transformation can be calculated as the sum of the first d eigenvalues divided by the sum of all eigenvalues.

$$C = X^T X / (n - 1), \quad (3.2)$$

$$\mathbf{C} = \mathbf{V}\mathbf{L}\mathbf{V}^\top, \quad (3.3)$$

$$\mathbf{Y} = \mathbf{X}\mathbf{V}, \quad (3.4)$$

However, in this case, an SVD-based implementation of PCA has been used. If SVD is applied on $\mathbf{X}$, it is obtained:

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top, \quad (3.5)$$

where $\mathbf{U}$ is a orthonormal matrix and $\mathbf{S}$ is the diagonal matrix of singular values. Replacing in the covariance matrix:

$$\mathbf{C} = \mathbf{V}\mathbf{S}\mathbf{U}^\top\mathbf{U}\mathbf{S}\mathbf{V}^\top / (n - 1) = \mathbf{V} \frac{\mathbf{S}^2}{n - 1} \mathbf{V}^\top, \quad (3.6)$$

which means that the right matrix of singular values ($\mathbf{V}$) are the eigenvectors and principal directions, and eigenvalues are related with the singular values ($\mathbf{S}$) [Shlens, 2014]. The principal components, i.e. the projection of the data in their new space is given by:

$$\mathbf{X}\mathbf{V} = \mathbf{U}\mathbf{S}\mathbf{V}^\top\mathbf{V} = \mathbf{U}\mathbf{S} \quad (3.7)$$

During the experiments, the hyperparameter d of the projection dimensions will be trained in order to obtain the optimal dimensional space in which to map the data.